A Three-Phase Energy Analyzer Node on the ESP32: Firmware DSP Core and Host Validation

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Abstract

This paper presents the measurement core of a three-phase energy analyzer intended to run on an ESP32 and publish its results over Modbus TCP. From sampled voltage and current the firmware computes true RMS values, active, apparent and non-active power, the true power factor, and the total harmonic distortion (THD) of each phase. A ten-cycle sampling window at 256 samples per cycle places every harmonic on an exact DFT bin, so harmonic magnitudes are obtained by per-order correlation rather than a full FFT. The same C core is compiled on a host computer and driven with synthetic waveforms of known harmonic content; it reproduces an independent numpy reference to a maximum relative error of 6.7×10^{-5}, and the measured THD matches the injected spectrum analytically (3.08% voltage, 9.90% current).

1 Introduction

Energy and power-quality monitoring at the feeder level relies on devices that sample the mains and reduce each window to a handful of quantities: RMS, power, power factor and distortion. Standard definitions for these quantities under non-sinusoidal conditions are given in IEEE Std 1459 (IEEE , 2010), while the measurement methods—aggregation windows and harmonic evaluation—are set by IEC 61000-4-30 (IEC , 2015) and IEC 61000-4-7 (IEC , 2002). The aim here is to implement that core on a low-cost microcontroller with a footprint small enough for real-time operation, and to show that the implementation is numerically correct before any hardware bring-up. The harmonic step uses a single-bin evaluation in the spirit of the Goertzel algorithm (Goertzel , 1958), which costs far less than a full transform when only a few orders are needed.

2 Method

2.1 Sampling window

The window spans ten mains cycles at 256 samples per cycle, i.e. $N = 2560$ samples and a per-channel rate of 12.8 kHz. A ten-cycle (200 ms at 50 Hz) aggregation is the basic measurement interval of IEC 61000-4-30 (IEC , 2015). Because the window is an integer number of cycles, each harmonic falls exactly on a DFT bin and no window function is required.

2.2 RMS and power

With $v[k]$ and $i[k]$ the per-sample voltage and current in engineering units,

$$V_{rms} = \sqrt{\tfrac{1}{N}\textstyle\sum_k v[k]^2}, \qquad I_{rms} = \sqrt{\tfrac{1}{N}\textstyle\sum_k i[k]^2}, \tag{1}$$

$$P = \tfrac{1}{N}\sum_k v[k]\,i[k], \qquad S = V_{rms}I_{rms}, \qquad Q = \sqrt{S^2 - P^2}, \qquad PF = \frac{P}{S}. \tag{2}$$

Here PF is the *true* power factor; under distortion it falls below the fundamental displacement factor $\cos\varphi_1$, consistent with IEEE Std 1459 (IEEE , 2010).

2.3 Harmonic magnitudes and THD

For harmonic order h the peak amplitude is obtained from the correlation of the signal with the quadrature pair at that order,

$$A_h = \frac{2}{N}\sqrt{\Big(\sum_{k} x[k]\cos\tfrac{2\pi hk}{s}\Big)^2 + \Big(\sum_{k} x[k]\sin\tfrac{2\pi hk}{s}\Big)^2}, \tag{3}$$

with $s = 256$ samples per cycle, and the total harmonic distortion is

$$\mathrm{THD} = \frac{\sqrt{\sum_{h=2}^{15} A_h^2}}{A_1}. \tag{4}$$

Evaluating only the needed orders keeps the cost linear in the number of harmonics, which suits the microcontroller target (Goertzel , 1958).

2.4 Validation setup

The firmware DSP source (`power_metrics.c`) is compiled unchanged on the host and fed a synthetic three-phase record: 230 V phase-neutral voltage carrying 3rd (1.5%), 5th (2.5%) and 7th (1.0%) harmonics, and 20 A current at a fundamental power factor of 0.90 lagging with 5th (8%), 7th (5%) and 11th (3%) current harmonics. An independent numpy implementation of the same definitions provides the reference. Figure 1 shows the phase-A waveforms and Figure 2 the current spectrum.

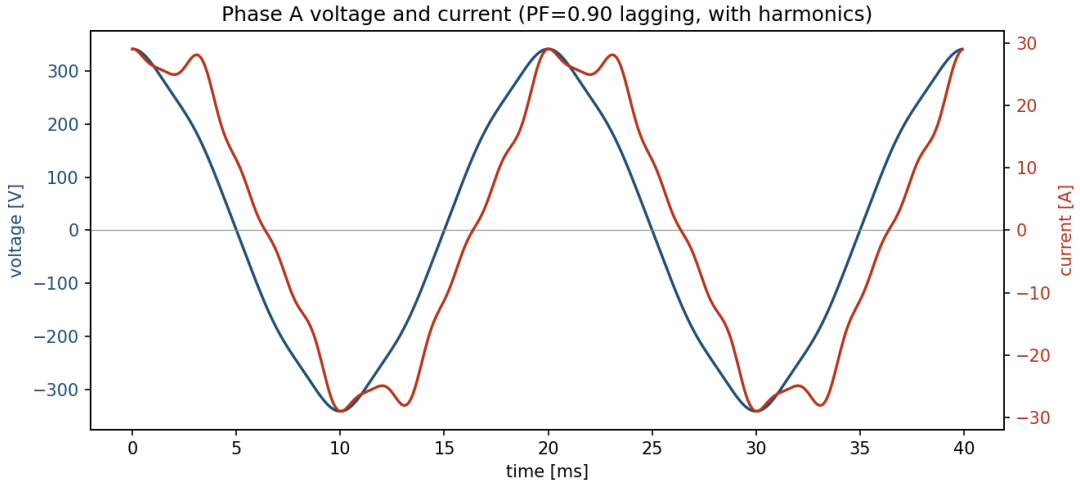


Figure 1: Synthetic phase-A voltage and current (0.90 lagging, with harmonics).

3 Results

Table 1 lists the firmware output against the numpy reference for phase A. All quantities agree to at least five significant figures; the largest relative error is 6.7×10^{-5}, on the THD values, and traces to the firmware using single precision while the reference uses double. The measured THD equals the injected content exactly ($\mathrm{THD}_v = 3.082\%$, $\mathrm{THD}_i = 9.899\%$), confirming that the

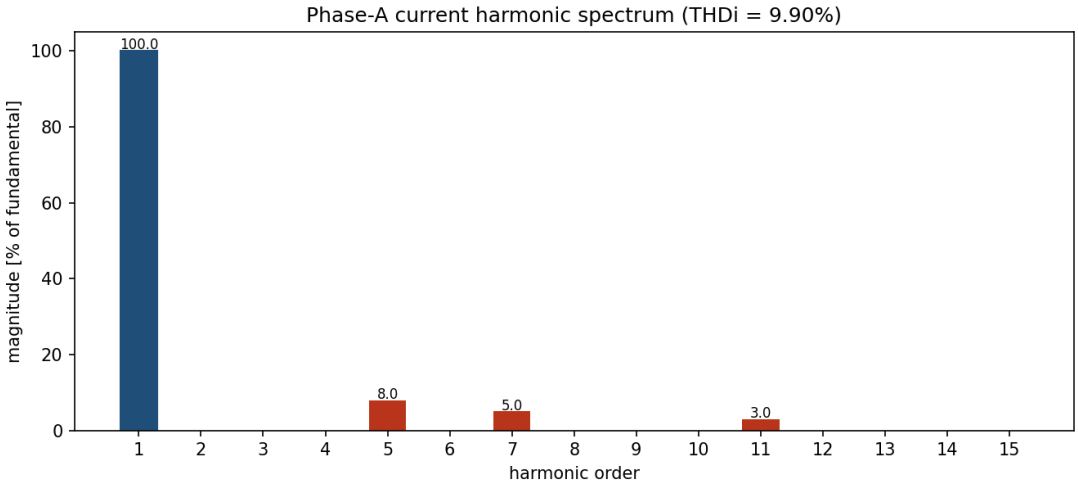


Figure 2: Phase-A current harmonic spectrum; the 5th, 7th and 11th orders dominate the distortion.

Table 1: Firmware DSP versus independent numpy reference (phase A).

Quantity	Firmware (C)	Reference (numpy)	Rel. error
V_{rms} [V]	230.1092	230.1092	1.1×10^{-7}
I_{rms} [A]	20.0978	20.0978	1.9×10^{-6}
P [W]	4149.711	4149.712	1.3×10^{-7}
S [VA]	4624.680	4624.680	4.4×10^{-8}
PF	0.89730	0.89730	3.4×10^{-6}
THD_v	0.03082	0.03082	6.7×10^{-5}
THD_i	0.09899	0.09899	5.0×10^{-5}

harmonic evaluation and the summation are correct. Figure 3 plots the per-metric relative error, all well inside a 0.01% band.

The true power factor comes out as 0.897, below the 0.90 fundamental displacement factor set in the test; the 9.9% current distortion accounts for the difference, as expected from the true-RMS definition. The three-phase active power is 12.45 kW.

4 Conclusion

The measurement core of a three-phase energy analyzer was implemented for the ESP32 and verified by compiling the identical C source on a host and comparing it against an independent reference over synthetic waveforms with known harmonics. Agreement to 6.7×10^{-5} and the exact recovery of the injected THD give confidence that the algorithm is correct ahead of hardware bring-up. The remaining work—ADC timing, front-end calibration and field verification against a reference meter—is hardware-specific and left for commissioning.

References

Goertzel, G. (1958). An algorithm for the evaluation of finite trigonometric series. *The American Mathematical Monthly, 65*(1), 34–35. https://doi.org/10.2307/2310304

International Electrotechnical Commission. (2002). *IEC 61000-4-7 Electromagnetic compatibility*

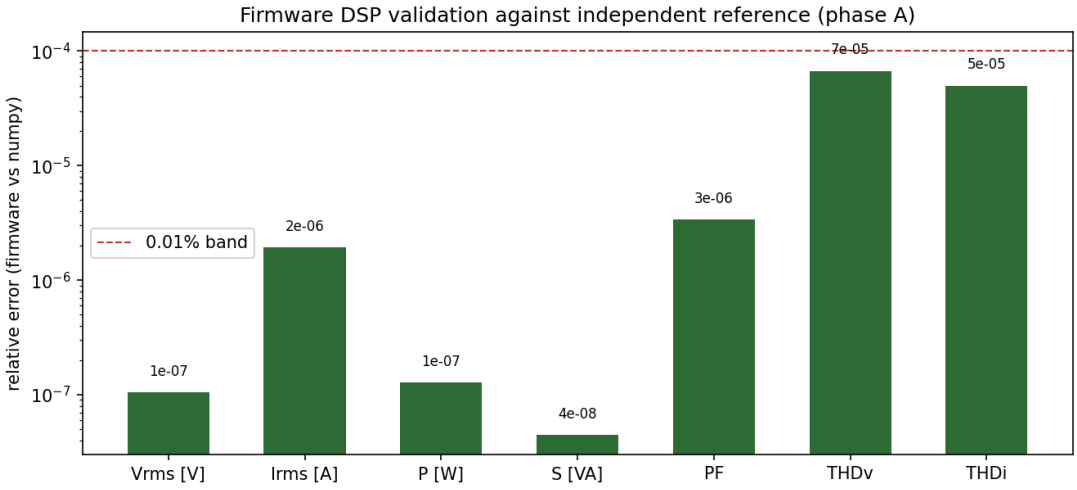


Figure 3: Per-metric relative error of the firmware core against the reference (log scale).

(EMC) – Part 4-7: Testing and measurement techniques – General guide on harmonics and interharmonics measurements and instrumentation. IEC.

International Electrotechnical Commission. (2015). *IEC 61000-4-30 Electromagnetic compatibility (EMC) – Part 4-30: Testing and measurement techniques – Power quality measurement methods*. IEC.

Institute of Electrical and Electronics Engineers. (2010). *IEEE Std 1459-2010 – IEEE standard definitions for the measurement of electric power quantities under sinusoidal, nonsinusoidal, balanced, or unbalanced conditions*. IEEE. https://doi.org/10.1109/IEEESTD.2010.5439063